Name: _____

Class: _____

Mr. Rawson • GCSE Computer Science Year 10 2026

CPU Performance Lab: Cores and Cache

How the lab works. Work in a group of four. **You are the CPU.** One person is the timekeeper: time every run in seconds on the class clock and write it in the table. Each run says who does what.

Lab 1 More cores, more work at the same time

Job A, a parallel job: eight sums that do not depend on each other, so any person can do any sum and they can all happen at once. There is one set of sums for Run 1 and another for Run 2.

Run 1 sums			Run 2 sums		
1	47 + 38	85	1	48 + 85	133
2	56 + 27	83	2	53 + 69	122
3	83 + 19	102	3	77 + 46	123
4	64 + 58	122	4	38 + 94	132
5	35 + 49	84	5	66 + 27	93
6	72 + 68	140	6	59 + 74	133
7	29 + 76	105	7	85 + 39	124
8	91 + 37	128	8	43 + 78	121

Job B, a sequential job: one chain of five steps. Each step uses the answer from the step before, so the steps happen in order. Write each answer in its box.

Run 3 chain	Start 4	×3	+17	×2	−13	+28
Answers		12	29	58	45	73
Run 4 chain	Start 5	×4	+16	×2	−9	+37
Answers		20	36	72	63	100

Run	Cores	Who does what	Time (s)
1	1	One person does all eight Run 1 sums. The other three watch.	≈ 60
2	4	Each person takes two of the Run 2 sums. All four work at the same time.	≈ 18
3	1	One person does the whole Run 3 chain.	≈ 12
4	4	Four people share the Run 4 chain: each does one step, then passes the answer to the next person.	≈ 12

Worked example

Worked example. Suppose Run 1 took 60 s and Run 2 took 20 s.
Speed-up = time on 1 core ÷ time on 4 cores = 60 ÷ 20 = 3.
Four people made the job three times faster.

Job A speed-up (Run 1 ÷ Run 2): . . . . . **Job B speed-up** (Run 3 ÷ Run 4): . . . . . .

Check your result. A speed-up close to **4** means the job split perfectly across the cores. A speed-up close to **1** means the extra cores had nothing to do. Your two speed-ups should be far apart.

Lab 2 A bigger cache means fewer trips to RAM

Your task: find the total cost of an order. Prices are in pence. The full price list is the **RAM**: it holds every item, and each look-up means finding the item in the list below.

RAM: the price list (pence)							
M	65	F	15	A	45	P	10
K	90	C	55	H	35	B	30
O	85	E	80	D	25	J	20
G	60	N	50	I	70	L	40

Run 5: no cache. One person looks up **every** price in the RAM list, one item at a time, and adds the running total.

Order X: A B A C A B D A C B A B **Total:** . . . . . pence (480) **Run 5 time (s):** . . . . . (≈ 50)

Run 6: with a cache. The order below uses only four different items. Before you start, look through the order and fill in the cache box: the **four items**, the **price** of each from the RAM list, and **how many times each is bought**. Then add up the order using **only the cache box**.

Cache	B	A	C	D
Item				
Price (pence)	30	45	55	25
Times bought	3	4	3	2

Order Y: B A C D A B C A D C B A **Total:** . . . . . pence (485) **Run 6 time (s):** . . . . . (≈ 25)

Check your result. Run 6 took less time than Run 5, and your two totals are correct.

What the lab shows

Complete each sentence. Use words from the lab.

1. Splitting Job A across four cores made it . . . . . about 3–4 times faster, because the sums . . . . . could all be done at the same time

2. Job B gained almost nothing from four cores, because . . . . . each step needs the answer from the step before, so the steps must run in order

3. Run 6 was faster than Run 5 because the cache held . . . . . the prices used most often , so there were fewer trips to . . . . . RAM , which is slower.

4. If every person in the group had worked twice as fast, every run would take about half as long. That is what a higher . . . . . clock speed does for a CPU.

Stretch. A job takes **12 minutes** on one core. **8 minutes** of it can be split perfectly across any number of cores. The other **4 minutes** must run on one core, in order.

(a) How long does the job take on **4 cores**? . . . . . 8 ÷ 4 + 4 = 6 minutes

(b) How long does it take on **8 cores**? . . . . . 8 ÷ 8 + 4 = 5 minutes

(c) Going from 4 cores to 8 cores doubled the cores. By how much did the time fall, and what sets the shortest time this job could ever reach?

. . . . . Only from 6 to 5 minutes. The 4 minutes that cannot be split set a floor of 4 minutes.

Teacher key. The times shown are typical; expect Run 1 about 45 to 90 s and Run 2 about a third to a quarter of that, with Run 3 and Run 4 within a couple of seconds of each other. If a group's Run 4 is *slower* than Run 3, that is the passing of the answers, and it makes the same point. For Run 5 and Run 6, expect the cache run to take roughly half the time once four prices are copied. Lab 2's totals: Order X = 5 A + 4 B + 2 C + 1 D = 225 + 120 + 110 + 25 = 480 p; Order Y = 4 A + 3 B + 3 C + 2 D = 180 + 90 + 165 + 50 = 485 p. Times bought in Order Y: A 4, B 3, C 3, D 2.